

Direction-of-Arrival Tracking Loop for Single-Vector Hydrophone

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Abstract—In dynamic scenarios, traditional single-vector hydrophone DOA estimation methods suffer from low accuracy. In this study, for the first time, the concept of a single-vector hydrophone DOA tracking loop is introduced. It employs an average intensity estimator or an active average intensity estimator as the phase discriminator and utilizes a second-order phase-locked loop to dynamically track the incoming direction of the signal. For the optimal synthesis of the velocity, the output of the phase discriminator in the phase-locked loop is utilized. Computer simulation and analysis confirm the effectiveness of this approach, revealing accurate DOA tracking and optimal velocity channel combining in dynamic environments.

Keywords—vector hydrophone, direction-of-arrival, phase-locked loop

I. INTRODUCTION

The complexity and variability of the underwater acoustic channel show significant challenges for underwater communication and measurement. A single-vector hydrophone (SVH) comprises an omnidirectional pressure sensor and dipole-directional particle velocity (acceleration) sensors. It can simultaneously measure the sound pressure and particle velocity at a specific point in the sound field. A single SVH can estimate the direction of arrival (DOA) of a sound source and provide good spatial directivity. By utilizing the DOA estimation from the signal, the pressure and velocity outputs of the vector hydrophone can be combined by proper weights to achieve spatial gains of 4.8 to 6 dB in an isotropic noise field [1].

Several options for DOA estimation are found in the literature [2, 3], including average intensity estimator, complex intensity estimator, and active average intensity estimator. The frequency-domain acoustic intensity method, also known as the cross-spectrum method, represents the time-domain acoustic intensity in the frequency domain. Its advantage lies in the ability to estimate the DOA of multiple targets with distinct spectral characteristics within a given bandwidth. It performs well in estimating the DOA of wideband signals such as ship-radiated noise. Yin [4] proposed an active average intensity estimator algorithm for CDMA underwater communication with an SVH. By utilizing the excellent autocorrelation and

cross-correlation properties of the pseudo-random sequence code, the algorithm can simultaneously estimate the DOAs of multiple users in the same frequency band, enabling directional communication and increasing the number of users.

The aforementioned SVH DOA estimation methods are based on batch processing, where DOA estimation is performed in segments and then electronic rotation is utilized to form velocity combinations. These static estimation methods do not utilize historical data and cannot achieve continuous tracking. In practical applications, when the receiving end rotates on the water surface or there is relative motion between the transmitter and receiver, the DOA of the communication target will change over time, which limits the performance of DOA estimation methods and results in low accuracy in dynamic scenarios. To address this issue, some algorithms have been proposed, like Kalman filtering and particle filtering [5]. However, these algorithms contain matrix operations and are too complicated to be implemented in real time.

This study proposes an SVH DOA estimation method based on a phase-locked loop (PLL). It utilizes an average intensity estimator or active average intensity estimator as the phase discriminator and employs a second-order PLL to dynamically track the incoming direction of the signal. In the meantime, the in-phase branch of the phase discriminator outputs the optimal combination of velocity channels. This method has low computational complexity and can track the signal's direction in real time.

II. MEASUREMENT MODEL

Under far-field conditions, neglecting the differences in sensitivity between the pressure and velocity channels of the vector sensor, the output of a two-dimensional SVH can be represented in the following form:

$$\begin{cases} p(t) = x(t) + n_p(t) \\ v_x(t) = x(t)\cos\theta + n_x(t) \\ v_y(t) = x(t)\sin\theta + n_y(t) \end{cases} \quad (1)$$

where

- $p(t)$ represents the received sound pressure signal by the SVH.
- $x(t) = s(t) \otimes h(t)$ represents the incoming signal, where $s(t)$ is the transmitted signal from the source, $h(t)$ is the channel impulse response, and $\otimes$ stands for time convolution.
- $v_x(t)$ and $v_y(t)$ represent the received velocity signals along the x -axis and y -axis, respectively.
- θ denotes the azimuth angle of the signal arrival direction, with 0 degrees defined as the positive x -axis.
- $n_p(t)$, $n_x(t)$, and $n_y(t)$ are mutually independent white noises with variances of σ_p^2 , σ_{vx}^2 , and σ_{vy}^2 , respectively, satisfying the following relationships [6]:

$$\sigma_p^2 = 2\sigma_{vx}^2 = 2\sigma_{vy}^2 = \sigma_n^2. \quad (2)$$

In (2), σ_n^2 is defined as the power of environmental noise, can be modeled as follows:

$$\sigma_n^2 = BN_0 = f_{\text{samp}}N_0/2, \#(3)$$

where N_0 represents the noise power spectral density, f_{samp} represents the sampling frequency of the hydrophone, and B represents the bandwidth of the noise.

Therefore, when only isotropic noise is present, the covariance matrix of the three channels of the SVH is given by [6]

$$\begin{aligned} \mathbf{R}_n &= \begin{bmatrix} \text{var}(p) & \text{cov}(p, v_x) & \text{cov}(p, v_y) \\ \text{cov}(v_x, p) & \text{var}(v_x) & \text{cov}(v_x, v_y) \\ \text{cov}(v_y, p) & \text{cov}(v_y, v_x) & \text{var}(v_y) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \sigma_n^2. \end{aligned} \quad (4)$$

When the SVH receives signals without any noise, the signals of the three channels can be represented in matrix form as follows:

$$\mathbf{X}(t) = \begin{bmatrix} p(t) \\ v_x(t) \\ v_y(t) \end{bmatrix} = \begin{bmatrix} 1 \\ \cos\theta \\ \sin\theta \end{bmatrix} x(t) = \mathbf{A}(\theta)x(t). \#(5)$$

In the equation, $\mathbf{A}(\theta)$ represents the signal steering vector, is given as follows:

$$\mathbf{A}(\theta) = \begin{bmatrix} 1 \\ \cos\theta \\ \sin\theta \end{bmatrix}. \quad (6)$$

Therefore, when the SVH receives signals without any noise, the covariance matrix of the three channels of the SVH is given by [6]

$$\begin{aligned} \mathbf{R}_s &= \frac{1}{T} \int_T \mathbf{X}(t)\mathbf{X}^T(t)dt \\ &= \frac{1}{T} \int_T \mathbf{A}(\theta)\mathbf{A}^T(\theta)x(t)x(t)dt \\ &= \mathbf{A}(\theta)\mathbf{A}^T(\theta) \cdot \frac{1}{T} \int_T x(t)x(t)dt \\ &= \begin{bmatrix} 1 & \cos\theta & \sin\theta \\ \cos\theta & \cos^2\theta & \cos\theta\sin\theta \\ \sin\theta & \sin\theta\cos\theta & \sin^2\theta \end{bmatrix} \sigma_s^2. \#(7) \end{aligned}$$

In (7), σ_s^2 represents the power of the signal received by the hydrophone, and T represents the integration time.

$$\sigma_s^2 = \frac{1}{T} \int_T x(t)x(t)dt \#(8)$$

III. ALGORITHM

Unlike the currently open-loop direction estimation algorithms, the method proposed in this study is a closed-loop estimation algorithm, as shown in Fig. 1. The algorithm comprises electronic rotation, cross-correlation, phase discriminator, loop filter, and numerically controlled oscillator (NCO), forming a phase-locked loop. This approach not only obtains the estimated azimuth angle $\hat{\theta}$ but also outputs the optimized velocity combinations, which are v_c and v_s .

First, the velocity vector (v_x, v_y) undergoes electronic rotation, with a rotation angle of $\hat{\theta}$, which is outputted by NCO. Then, an open-loop estimation of the error quantity $(\theta - \hat{\theta})$ is performed. The tracking loop in this study is compatible with various DOA open-loop estimation algorithms. These open-loop estimation algorithms are utilized to estimate $(\theta - \hat{\theta})$, and the estimated value of $(\theta - \hat{\theta})$ serves as the output of the phase discriminator. After passing through the loop filter, it controls the NCO to generate $\hat{\theta}$. The phase-locked loop is a negative feedback loop that iteratively approaches 0 for $(\theta - \hat{\theta})$.

For simplicity, this study adopts the average intensity estimator and active average intensity estimator as the open-loop estimator. The following sections explain each module of the algorithm.

A. Electronic Rotation

Based on the estimated azimuth angle $\hat{\theta}$, the electronic rotation of v_x and v_y yields v_s and v_c . The electronic rotation can be expressed using complex notation as follows:

$$v_c + i \cdot v_s = (v_x + i \cdot v_y)e^{-i\hat{\theta}}. \#(9)$$

The real and imaginary parts can be expressed as follows:

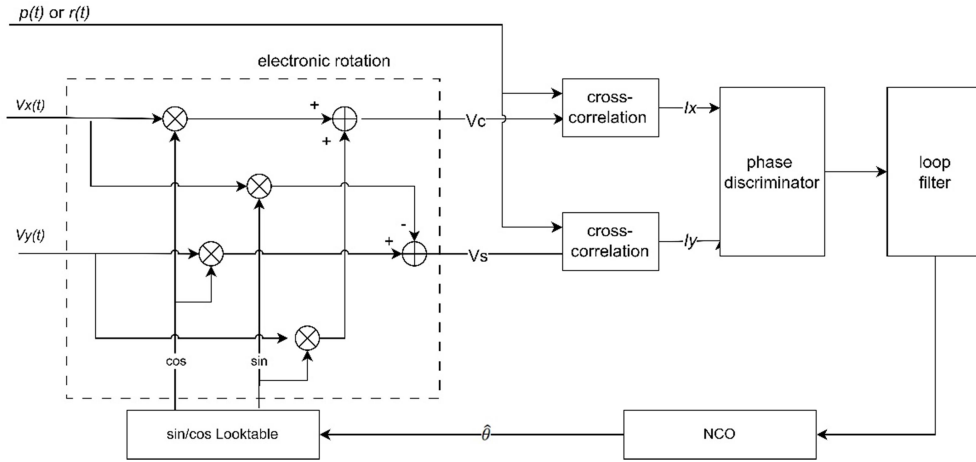


Fig. 1. Block diagram of DOA tracking loop.

$$\begin{aligned}
 v_c &= v_x \cos \hat{\theta} + v_y \sin \hat{\theta} \\
 &= x(t) \cos(\theta - \hat{\theta}) + (n_x \cos \hat{\theta} + n_y \sin \hat{\theta}) \\
 &= x(t) \cos(\theta_e) + n_c,
 \end{aligned} \quad (10)$$

$$\begin{aligned}
 v_s &= -v_x \sin \hat{\theta} + v_y \cos \hat{\theta} \\
 &= x(t) \sin(\theta - \hat{\theta}) + (-n_x \sin \hat{\theta} + n_y \cos \hat{\theta}) \\
 &= x(t) \sin(\theta_e) \\
 &\quad + n_s,
 \end{aligned} \quad (11)$$

where $\theta_e = \theta - \hat{\theta}$ and n_c, n_s symbolize the noise components in the v_c and v_s .

B. Cross-Correlation

The simplest method for DOA estimation using SVH is the average intensity estimator. In this method, the received pressure signal $p(t)$ is cross-correlated with v_c and v_s , resulting in I_x and I_y , respectively.

$$I_x = \int_0^T p(t) v_c(t) dt = T \sigma_s^2 \cos \theta_e + \Delta_x, \quad (12)$$

$$I_y = \int_0^T p(t) v_s(t) dt = T \sigma_s^2 \sin \theta_e + \Delta_y, \quad (13)$$

where T stands for the cross-correlation integration time. On the right side of (12) and (13), the first term is the signal component, and the second term is the noise component.

If the receiver has knowledge of the reference waveform $r(t)$ of the received signal, such as the spreading code in CDMA communication, the active average intensity estimator can be employed. In this method, the reference waveform $r(t)$ is cross-correlated with v_c and v_s , resulting in I_x and I_y , respectively.

$$I_x = \int_0^T r(t) v_c(t) dt = T \sigma_s^2 \cos \theta_e + \Delta_x, \quad (14)$$

$$I_y = \int_0^T r(t) v_s(t) dt = T \sigma_s^2 \sin \theta_e + \Delta_y. \quad (15)$$

C. Phase Discriminator

Using the inverse tangent (arctan) function to obtain an estimation of θ_e :

$$\hat{\theta}_e = \tan^{-1}(I_y, I_x), \quad (16)$$

where $\tan^{-1}(\cdot)$ is a four-quadrant inverse tangent function and $\hat{\theta}_e \in (-\pi, \pi)$.

D. Loop Filter

In the context of oceanic applications, where dynamics are relatively small when compared with aerospace applications, a second-order PLL should be sufficient to meet the dynamic tracking requirements of underwater acoustics. In this case, the loop filter is implemented as a first-order infinite impulse response filter, as shown in Fig. 2. The input to the loop filter is $\hat{\theta}_e$ (estimated error), and the output is the frequency control signal. The transfer function $F(z)$ of the loop filter can be given as follows [7]:

$$F(z) = \frac{(C_1 + C_2) - C_1 z^{-1}}{1 - z^{-1}}. \quad (17)$$

The loop filter has well-established design methods, where C_1 and C_2 are calculated based on the loop parameters (B_L, ζ, k), where B_L is the loop bandwidth, ζ is the damping ratio, and k is the loop gain. The natural frequency is calculated from the loop parameters according to the following [7]:

$$\omega_n = \frac{8\zeta B_L}{4\zeta^2 + 1}. \quad (18)$$

Then, the values of C_1 and C_2 for the filter can be calculated according to the following [7]:

$$C_1 = \frac{1}{k} \frac{8\zeta \omega_n T}{4 + 4\zeta \omega_n T + (\omega_n T)^2}, \quad (19)$$

$$C_2 = \frac{1}{k} \frac{4(\omega_n T)^2}{4 + 4\zeta \omega_n T + (\omega_n T)^2}. \quad (20)$$

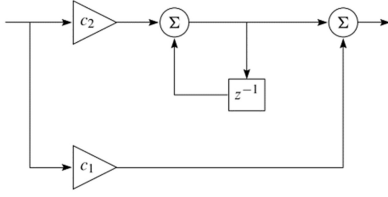


Fig. 2. Second-order phase lock loop filter.

E. NCO

The output of the loop filter is the frequency control signal $f(t)$, which serves as the input to the NCO. The NCO generates the phase output $\theta(t)$, implementing the integral operation:

$$\theta(t) = \int_0^t 2\pi \cdot f(t) dt. \quad (21)$$

The phase output $\theta(t)$ corresponds to the estimated azimuth angle $\hat{\theta}$ mentioned above, which is a function of time.

IV. SIMULATION AND ANALYSIS

To investigate the performance of the tracking loop described in this study and the impact of different parameters, we use computer simulations. The simulations employ a phase discriminator based on the average intensity estimator.

The main parameter settings are as follows: sampling frequency is 48 kHz, carrier frequency is 12 kHz, signal waveform is linear frequency modulation (LFM), signal and noise bandwidths are 8 kHz, and simulation durations are 200 seconds. The signal-to-noise ratio (SNR), defined as $SNR = 10\lg(\sigma_s^2/\sigma_n^2)$, ranges from -10 to 10 dB in 1-dB increments. The simulations include three scenarios: stationary, circular motion, and linear motion.

A. Scenario 1

The source and hydrophone remain stationary, with azimuth angle $\theta = 60^\circ$. In the simulation, we first set the cross-correlation integration time $T = 0.1$ s and vary the loop bandwidth B_L to values of 0.1, 0.2, 0.3, 0.4, and 0.5 Hz. The simulation results in different SNRs are used to calculate the root mean square error (RMSE) of the azimuth estimation, as shown in Fig. 3. Fig. 4 shows the comparison between the estimated DOA and the true value of θ .

In Fig. 3, as the loop bandwidth B_L decreases, the RMSE decreases, which indicates a higher estimation accuracy of the algorithm. When $SNR > 0$ dB, the RMSEs of the DOA tracking loop with different B_L values are less than 1° , demonstrating high precision in tracking and estimation.

Fig. 5 displays the RMSEs of the azimuth angle estimation with $B_L = 0.1$ Hz versus the cross-correlation integration time, including 0.1, 0.2, 0.5, 0.8, and 1 s. The figure shows that shorter integration times have a slight advantage in performance. However, the performance difference among different integration times is relatively small. Therefore, for subsequent simulations, $T = 0.1$ s is chosen as the cross-correlation integration time.

B. Scenario 2

The source moves in a uniform circular motion around the hydrophone, completing one full circle every 20 s. The simulation results are shown in Figs. 6, 7, and 8. Fig. 6 depicts the RMSE of the azimuth angle estimation for different SNRs, with the cross-correlation integration time $T = 0.1$ s and varying loop bandwidth values of $B_L = 0.1, 0.2, 0.3, 0.4$, and 0.5 Hz. Similar to the results in Scenario 1, smaller loop bandwidth (B_L) leads to smaller RMSE, which indicates a higher estimation accuracy of the algorithm. When $SNR > 0$ dB, the RMSEs of the DOA tracking loop with different B_L values are less than 1° , also demonstrating high precision.

In this dynamic scenario, the performance of the tracking loop is similar to the static scenario, without significant estimation errors caused by the circular motion. This is because the second-order phase-locked loop not only eliminates constant tracking errors but also mitigates constant-speed tracking errors. Therefore, in the dynamic scenario of uniform circular motion, the proposed DOA tracking loop exhibits no dynamic errors.

Figs. 8 and Fig. 9 show the comparison between the estimated DOA and the true value with $SNR = 10$ dB or -10 dB, respectively, demonstrating a high degree of consistency between the estimated and true values. Fig. 7 displays the tracking errors for different loop bandwidths, which indicates that tracking loops with smaller loop bandwidths have the advantage of higher tracking accuracy but slower convergence speed.

C. Scenario 3

Fig. 10 shows a two-dimensional illustration of the scenario. The signal source (a ship) moves in a straight line at a constant speed, passing near the hydrophone. The hydrophone is positioned at the origin, and the angle between the source's trajectory and the hydrophone's x -axis is set to -60° . The variation of the azimuth angle θ over time is illustrated in Fig. 13. When the source is closest to the hydrophone (at time point 100 s), the azimuth angle changes rapidly, causing θ to decrease quickly. Fig. 13(a) not only provides the true values of the azimuth angle but also displays the estimated values from the tracking loops with different B_L values, with $SNR = 10$ dB. For comparison, Fig. 13(b) presents the estimation of DOA with $SNR = -10$ dB in Scenario 3. The simulation demonstrates that the tracking performance is fairly satisfactory.

Fig. 11 shows the RMSE for different SNRs with varying loop bandwidths in Scenario 3. There is no obvious difference between Fig. 11, Fig. 3, and Fig. 6.

However, Fig. 12 shows that the tracking loop with $B_L = 0.1$ Hz exhibits noticeable dynamic errors in the range of -0.6° to 0.6° during the time interval of 60 to 140 s. This interval corresponds to the period when the source ship is closest to the hydrophone and the azimuth angle changes rapidly. The dynamic errors originate from the acceleration of the azimuth angle rather than environmental noise. In comparison, tracking loops with larger loop bandwidths exhibit smaller dynamic errors. This is because a second-order phase-locked loop with a smaller loop bandwidth introduces greater

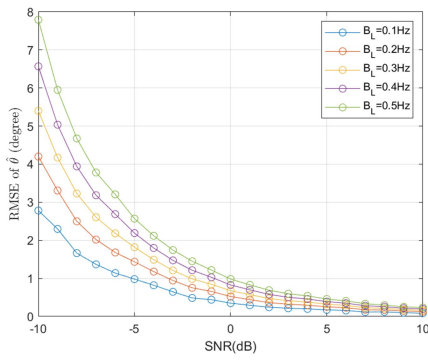


Fig. 3. RMSEs of the DOA tracking for different B_L , with $T = 0.1$ s in Scenario 1.

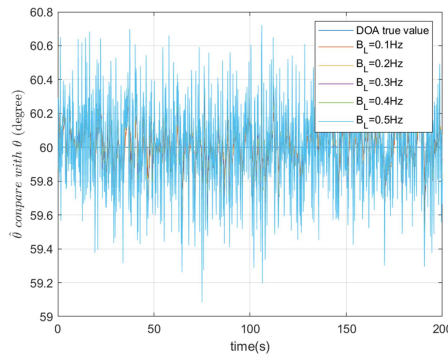


Fig. 4. DOA estimation compared with the true value for different B_L , with $T = 0.1$ s and $SNR = 10$ dB in Scenario 1.

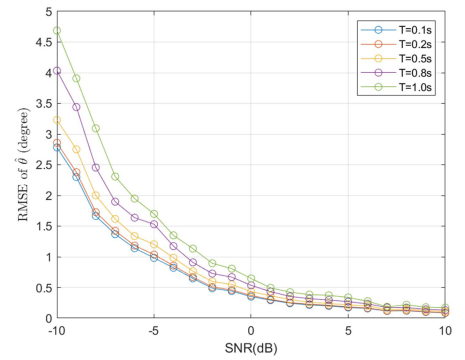


Fig. 5. RMSEs of the DOA tracking for different T , with $B_L = 0.1$ Hz in Scenario 1.

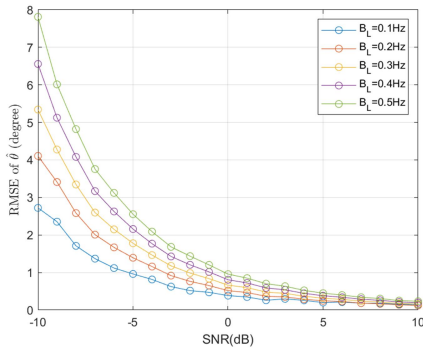


Fig. 6. RMSEs of the DOA tracking for different B_L , with $T = 0.1$ s in Scenario 2.

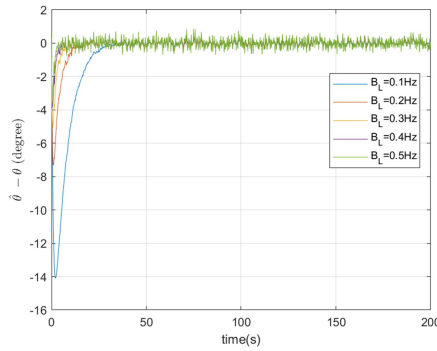


Fig. 7. Error of the DOA tracking for different B_L , with $SNR = 10$ dB, $T = 0.1$ s in Scenario 2.

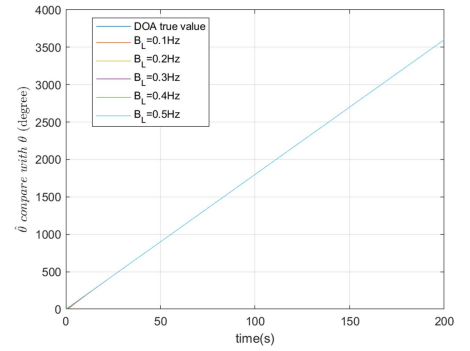


Fig. 8. Estimation of DOA compared with the true value for different B_L , with $T = 0.1$ s and $SNR = 10$ dB in Scenario 2.

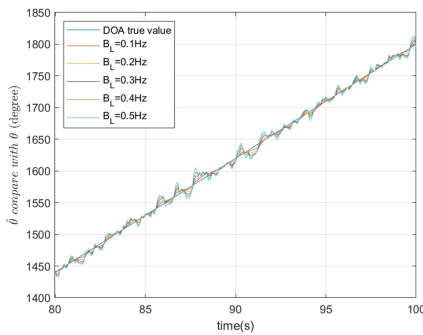


Fig. 9. Estimation of DOA compared with the true value for different B_L , with $T = 0.1$ s and $SNR = -10$ dB in Scenario 2.

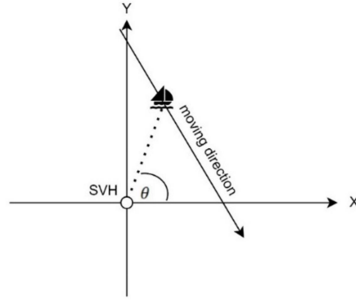


Fig. 10. Two-dimensional view of Scenario 3.

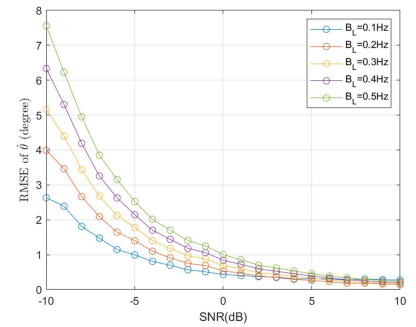


Fig. 11. RMSEs of the DOA tracking for different B_L , with $T = 0.1$ s in Scenario 3.

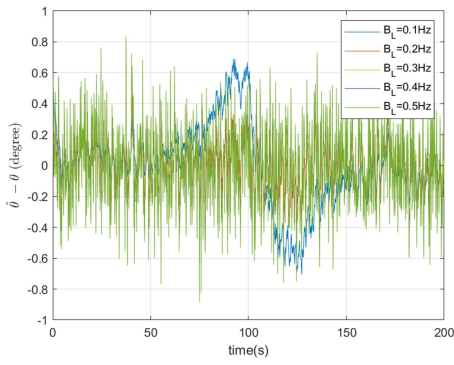


Fig. 12. Error of the DOA tracking for different B_L , with $SNR = 10$ dB, $T = 0.1$ s in Scenario 3.

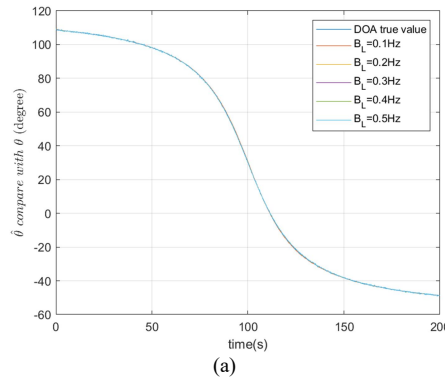
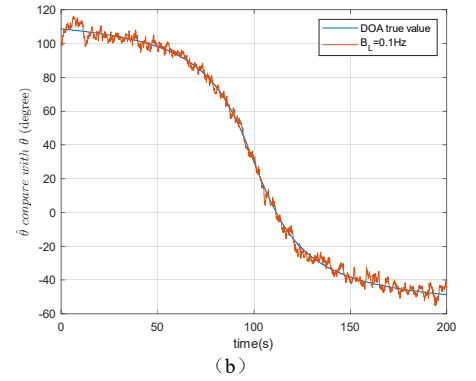


Fig. 13(a). Estimation of DOA compared with the true value, with $T = 0.1$ s, $SNR = 10$ dB, and different B_L in Scenario 3. (b) The estimation of DOA compare with the true value, with $B_L = 0.1$ Hz, $T = 0.1$ s, $SNR = -10$ dB in Scenario 3.



acceleration dynamic errors. Therefore, the choice of loop bandwidth must strike a balance. It cannot be arbitrarily reduced to improve DOA tracking accuracy. Tracking loops with very small loop bandwidths may have high accuracy but lower tolerance for acceleration dynamics and noise, which potentially leads to loss of lock.

V. CONCLUSION

This study proposes a DOA estimation method for SVH based on a PLL, specifically designed for underwater communication in dynamic scenarios. The method utilizes algorithms such as the average intensity estimator or the active average intensity estimator as the phase discriminator. It employs a second-order PLL to dynamically track the direction of incoming signals, iteratively approaching zero estimation error. When the estimation error approaches 0, the in-phase branch of the discriminator is the optimal combination of the velocity channel. Computer simulation results demonstrate that this method can achieve high-precision tracking of the incoming signal direction in various dynamic scenarios.

Notably, the tracking loop proposed in this study is an algorithm framework that can be compatible with various open-loop DOA estimation methods. It employs these open-loop estimation algorithms to implement the phase discriminator function within the loop.

Compared with tracking methods such as Kalman filtering and particle filtering, the proposed method has the advantage of lower computational complexity, making it more suitable for low-power underwater acoustic platforms.

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